Mitoprep Premium Test 10

Sapno Ki Udaan

Physics :- Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory of Gases, Oscillations, Waves

Chemistry :- Equilibrium, Redox Reactions, Organic Chemistry: Some Basic Principles & Techniques, Hydrocarbons

Biology :- Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth & Development, Breathing & Exchange of Gases, Body Fluids & Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

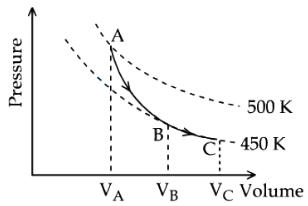
5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. 

A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A $\left(\mathrm{P_A} = 10^5 \text{ Pa}, \ \mathrm{V_A} = 4 \times 10^{-6} \text{ m}^3\right)$ to point B $\left(\mathrm{P_B} = 5 \times 10^4 \text{ Pa}, \ \mathrm{V_B} = 6 \times 10^{-6} \text{ m}^3\right)$ to point C $\left(\mathrm{P_C} = 10^4 \text{ Pa}, \mathrm{V_C} = 8 \times 10^{-6} \text{ m}^3\right)$. A to B is an adiabatic path and B to C is an isothermal path.
The net heat absorbed per unit mole by the system is :

A) $500\mathrm{R}(\ln 3 + \ln 4)$ **B)** $450\mathrm{R}(\ln 4 - \ln 3)$ **C)** $500\mathrm{R} \ln 2$ **D)** $400\mathrm{R} \ln 4$

2. In a stretched string,

A) Only transverse waves can exist
B) Only longitudinal waves can exist
C) Both transverse and longitudinal waves can exist
D) None of these

3. For a wave propagating in a medium, identity the property that is independent of the others

A) velocity
B) wavelength
C) frequency
D) all these depend on each other

4. A metal rod of $1 \ m$ length, is dropped exact vertically on to a hard metal floor. With an oscilloscope, it is determined that the impact produces a longitudinal wave of $1.2 \ kHz$ frequency. The speed of sound in the metal rod is

A) 2400 **B)** 1800 **C)** 1200 **D)** 600

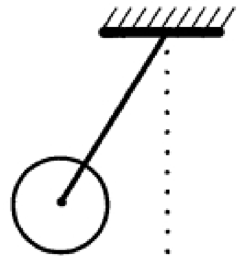
5. A wavelength $0.60 cm$ is produced in air and it travels at a speed of $300 ms^{-1}$. It will be an

A) Audible wave **B)** Infrasonic wave
C) Ultrasonic wave **D)** None of the above

6. Velocity of sound is maximum in

A) Air **B)** Water **C)** Vacuum **D)** Steel

7. A metal rod of length ' L ' and mass ' m ' is pivoted at one end. A thin disk of mass ' M ' and radius $'R'$ $(< L)$ is attached at its center to the free end of the rod. Consider two ways the disc is attached: (case A) The disc is not free to rotate about its center and (case B) the disc is free to rotate about its center. The rod-disc system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is (are) true?
$Image$
(A) Restoring torque in case A $=$ Restoring torque in case B
(B) Restoring torque in case A $<$ Restoring torque in case B
(C) Angular frequency for case A $>$ Angular frequency for case B.
(D) Angular frequency for case A $<$ Angular frequency for case B.



A) (A, D) **B)** (B, D)
C) (C, D) **D)** (A, B)

8. Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?

A) $a + bx - cx^2$ **B)** bx^2
C) $a - bx + cx^2$ **D)** $-bx$

9. $Assertion$: In SHM, acceleration is always directed towards the mean position.
$Reason$: In SHM, the body has to stop momentary at the extreme position and move back to mean position.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

C) If the Assertion is correct but Reason is incorrect.
D) If both the Assertion and Reason are incorrect.

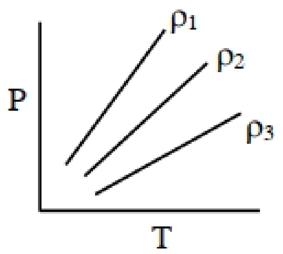
10. Which one of the following equations of motion represents simple harmonic motion ?
Where k, k_0, k_1 and a are all positive

A) $Acceleration = k(x)$
B) $Acceleration \ = k(x+a)$
C) $Acceleration \ = -k(x+a)$
D) $Acceleration \ = -k(x^2)$

11. What is constant in S.H.M.

A) Restoring force B) Kinetic energy
C) Potential energy D) Periodic time

12. $P - T$ diagram of an ideal gas having three different densities ρ_1, ρ_2, ρ_3 (in three different cases) is shown in the figure. Which of the following is correct :



A) $\rho_2 < \rho_3$ B) $\rho_1 > \rho_2$
C) $\rho_1 < \rho_2$ D) $\rho_1 = \rho_2 = \rho_3$

13. A cylinder contains hydrogen gas at pressure of $249\ kPa$ and temperature $27^\circ C$. Its density is $.............\ kg/m^3\ (R = 8.3\, J\, mol^{-1} K^{-1})$

A) 0.02 B) 0.5 C) 0.2 D) 0.1

14. A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T. The mass of each molecule of the gas is m. Which of the following gives the density of the gas ?

A) $\frac{P}{kTV}$ B) mKT
C) $\frac{P}{kT}$ D) $\frac{Pm}{kT}$

15. The translatory kinetic energy of a gas per gm is

A) $\frac{3}{2}\frac{RT}{N}$ B) $\frac{3}{2}\frac{RT}{M}$
C) $\frac{3}{2}RT$ D) $\frac{3}{2}NKT$

16. A balloon contains $1500\, m^3$ of helium at $27^\circ C$ and 4 atmospheric pressure. The value of helium at $-3^\circ C$ temperature and 2 atmospheric pressure will be ...... m^3

A) 1500 B) 1700 C) 1900 D) 2700

17. Match List I with List II :

List I	List II
A Isothermal Process	I Work done by the gas decreases internal energy
B Adiabatic Process	II No change in internal energy
C Isochoric Process	III The heat absorbed goes partly to increase internal energy and partly to do work
D Isobaric Process	IV No work is done on or by the gas

Choose the correct answer from the options given below :

A) $A-II, B-I, C-III, D-IV$
B) $A-II, B-I, C-IV, D-III$
C) $A-I, B-II, C-IV, D-III$
D) $A-I, B-II, C-III, D-IV$

18. A gas is compressed adiabatically, which one of the following statement is NOT true.

A) There is no heat supplied to the system
B) The temperature of the gas increases
C) The change in the internal energy is equal to the work done on the gas.
D) There is no change in the internal energy

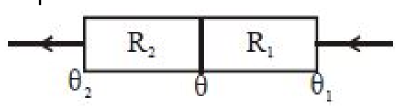
19. During the melting of a slab of ice at $273\ K$ at atmospheric pressure

A) Positive work is done by ice-water system on the atmosphere
B) Positive work is done on the ice-water system by the atmosphere
C) The internal energy of the ice-water system increases
D) (B) and (C) both

20. The state of a thermodynamic system is represented by

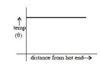
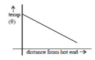
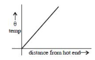
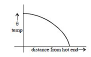
A) Pressure only
B) Volume only
C) Pressure, volume and temperature
D) Number of moles

21. Consider two insulating sheets with thermal resistances R_1 and R_2 as shown. The temperatures θ is

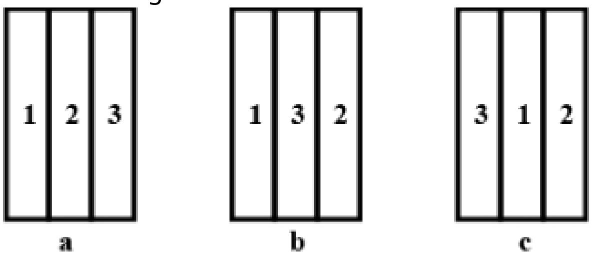


A) $\frac{\theta_1\theta_2 R_1 R_2}{(R_1+R_2)(\theta_1+\theta_2)}$ B) $\frac{\theta_1 R_1+\theta_2 R_2}{R_1+R_2}$
C) $\frac{(\theta_1+\theta_2)(R_1 R_2)}{R_1^2+R_2^2}$ D) $\frac{\theta_1 R_2+\theta_2 R_1}{R_1+R_2}$

22. In steady state graph between temp and distance from hot end is

A) 
B) 
C) 
D) 

23. Figure shows three different arrangements of materials $1, 2$ and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is $20\,^oC$ higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases



A) $\Delta T_a > \Delta T_b > \Delta T_c$
B) $\Delta T_a = \Delta T_b = \Delta T_c$
C) $\Delta T_a = \Delta T_b > \Delta T_c$
D) $\Delta T_a = \Delta T_b < \Delta T_c$

24. The ends of two rods of different materials with their thermal conductivities, radii of cross-sections and lengths all are in the ratio $1:2$ are maintained at the same temperature difference. If the rate of flow of heat in the larger rod is 4 $cal/\sec$, that in the shorter rod in $cal/\sec$ will be

A) 1 B) 2 C) 8 D) 16

25. Snow is more heat insulating than ice, because

A) Air is filled in porous of snow
B) Ice is more bad conductor than snow
C) Air is filled in porous of ice
D) Density of ice is more

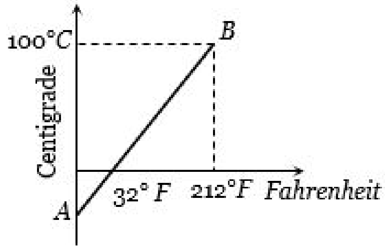
26. A piece of ice (heat capacity $= 2100\ \mathrm{J\ kg^{-1}{}^\circ C^{-1}}$ and latent heat $= 3.36 \times 10^5\ \mathrm{J\ kg^{-1}}$) of mass m grams is at $-5^\circ\mathrm{C}$ at atmospheric pressure. It is given $420\ \mathrm{J}$ of heat so that the ice starts melting. Finally when the ice-water mixture is in equilibrium, it is found that $1\ \mathrm{gm}$ of ice has melted. Assuming there is no other heat exchange in the process, the value of m is

A) 7 B) 8 C) 9 D) 5

27. If a bimetallic strip is heated, it will

A) Bend towards the metal with lower linear thermal expansion coefficient
B) Bend towards the metal with higher linear thermal expansion coefficient
C) Not bend at all
D) None

28. The graph AB shown in figure is a plot of temperature of a body in degree celsius and degree Fahrenheit. Then



A) Slope of line AB is $9/5$
B) Slope of line AB is $5/9$
C) Slope of line AB is $1/9$
D) Slope of line AB is $3/9$

29. The temperature of a body on Kelvin scale is found to be $x\ K$. When it is measured by Fahrenheit thermometer, it is found to be $x\,^\circ F$, then the value of x is

A) 40 B) 313 C) 574.25 D) 301.25

30. On centigrade scale the temperature of a body increases by 30 degrees. The increase in temperature on Fahrenheit scale is .... ^{o}

A) 50 B) 40 C) 30 D) 54

31. A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet)

A) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$ B) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$
C) $r = \sqrt{\frac{T}{(d-\rho)g}}$ D) $r = \sqrt{\frac{\mathrm{T}}{(\mathrm{d}+\rho)\mathrm{g}}}$

32. If T is the surface tension of soap solution, the amount of work done in blowing a soap bubble from a diameter D to $2D$ is

A) $2\pi D^2 T$ B) $4\pi D^2 T$ C) $6\pi D^2 T$ D) $8\pi D^2 T$

33. If the surface tension of a soap solution is 0.03 MKS units, then the excess of pressure inside a soap bubble of diameter $6\ mm$ over the atmospheric pressure will be

A) Less than $40\,N/m^2$
B) Greater than $40\,N/m^2$
C) Less than $20\,N/m^2$
D) Greater than $20\,N/m^2$

34. The surface tension for pure water in a capillary tube experiment is

A) $\frac{\rho g}{2hr}$ B) $\frac{2}{hr\rho g}$
C) $\frac{r\rho g}{2h}$ D) $\frac{hr\rho g}{2}$

35. A water drop takes the shape of a sphere in a oil while the oil drop spreads in water, because ($A.F.$ = adhesive force, $C.F.$ = cohesive force)

A) $C.F.$ for water $>$ $A.F.$ for water and oil
B) $C.F.$ for oil $>$ $A.F.$ for water and oil
C) $C.F.$ for oil $<$ $A.F.$ for water and oil
D) None of the above

36. Select wrong statement about pressure ....

A) Pressure is a scalar quantity
B) Pressure is always compressive in nature
C) Pressure at a point is same in all directions
D) None of these

37. By sucking through a straw, a student can reduce the pressure in his lungs to $750\, mm\, of\, Hg$ (density $= 13.6\, gm/cm^3$). Using the straw, he can drink water from a glass upto a maximum depth of .... cm

A) 10 B) 75 C) 13.6 D) 1.36

38. A cubical block of wood of specific gravity 0.5 and a chunk of concrete of specific gravity 2.5 are fastened together. The ratio of the mass of wood to the mass of concrete which makes the combination to float with its entire volume submerged under water is

A) $1/5$ B) $1/3$ C) $3/5$ D) $2/5$

39. During blood transfusion the needle is inserted in a vein where the gauge pressure is $2000\, Pa$. At what height (in m) must the blood container be placed so that blood may just enter the vein ? Density of whole blood, $\rho = 1.06 \times 10^3\ kgm^{-3}$

A) 0.1 B) 0.3 C) 0.2 D) 0.4

40. Two substances of densities ρ_1 and ρ_2 are mixed in equal volume and the relative density of mixture is 4. When they are mixed in equal masses, the relative density of the mixture is 3. The values of ρ_1 and ρ_2 are

A) $\rho_1 = 6$ and $\rho_2 = 2$
B) $\rho_1 = 3$ and $\rho_2 = 5$
C) $\rho_1 = 12$ and $\rho_2 = 4$
D) None of these

41. $Assertion$: Solids are least compressible and gases are most compressible.
$Reason$: solids have definite shape and volume but gases do not have either definite shape or definite volume.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
C) If the Assertion is correct but Reason is incorrect.
D) If both the Assertion and Reason are incorrect.

42. One end of a uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height $3L/4$ from its lower end is

A) $\frac{W_1}{S}$ B) $\frac{W_1+(W/4)}{S}$
C) $\frac{W_1+(3W/4)}{S}$ D) $\frac{W_1+W}{S}$

43. Modulus of rigidity of diamond is

A) Too less B) Greater than all matters
C) Less than all matters D) Zero

44. The length of an elastic string is a metre when the longitudinal tension is $4\, N$ and b metre when the longitudinal tension is $5\, N$. The length of the string in metre when the longitudinal tension is $9\, N$ is

A) $a - b$ B) $5b - 4a$
C) $2b - \frac{1}{4}a$ D) $4a - 3b$

45. A weight of $200\ kg$ is suspended by vertical wire of length $600.5\ cm$. The area of cross-section of wire is $1\ mm^2$. When the load is removed, the wire contracts by $0.5\, cm$. The Young's modulus of the material of wire will be

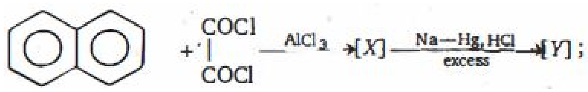
A) $2.35 \times 10^{12}\, N/m^2$ B) $1.35 \times 10^{10}\, N/m^2$
C) $13.5 \times 10^{11}\, N/m^2$ D) $23.5 \times 10^{9}\, N/m^2$

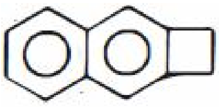
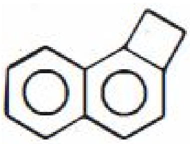
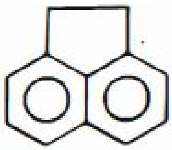
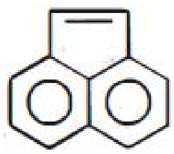
Chemistry

46. Which isomer of xylene can give three different monochloroderivatives ?

A) $o-$ xylene
B) $m-$ xylene
C) $p-$ xylene
D) xylene cannot give a monochloro derivative

47. Product Y is

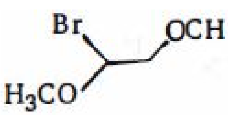


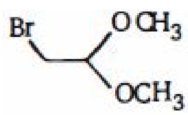
A) 
B) 
C) 
D) 

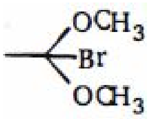
48. Trans-cyclohexane- $1,2$ -diol can be obtained by the reaction of cyclohexene with

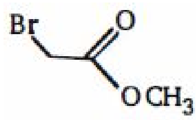
A) $KMnO_4$
B) OsO_4
C) peroxy formic acid / H_3O^+
D) SeO_2

49. Methyl vinyl ether, $H_2C=CH-OCH_3$, reacts with Br_2/CH_3OH. If methanol is reacting as water would, and if this reaction follows a typical mechanism of electrophilic addition, what would be the expected product ?

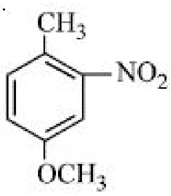
A) 

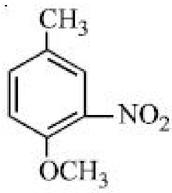
B) 

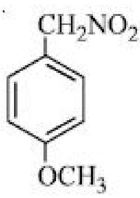
C) 

D) 

50. If $p-$ methoxy toluene is nitrated, the major product is :

A) 

B) 

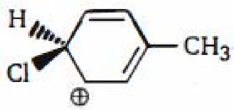
C) 

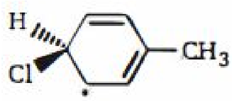
D) No reaction

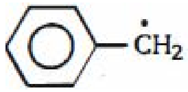
51. Which of the following has maximum stability

A) $\overset{+}{C}H_3$
B) $CH_3-\underset{+}{\overset{H|}{C}}-H$
C) $CH_3-\underset{+}{\overset{H|}{C}}-CH_3$
D) $CH_3-\underset{+}{\overset{CH_3|}{C}}-CH_3$

52. Which represents an intermediate formed in the reaction of toluene and chlorine at elevated temperature in sunlight ?

A) 

B) 

C) 

D) 

53. The correct order for acid strength of compounds $CH\equiv CH$, $CH_3-C\equiv CH$, $CH_2=CH_2$ is as follows

A) $CH\equiv CH>CH_2=CH_2>CH_3-C\equiv CH$
B) $CH_3-C\equiv CH>CH\equiv CH>CH_2=CH_2$
C) $CH_3-C\equiv CH>CH_2=CH_2>CH\equiv CH$
D) $CH\equiv CH>CH_3-C\equiv CH>CH_2=CH_2$

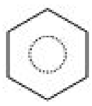
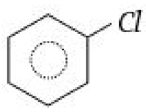
54. Which of the following molecules have non-zero dipole moments ?
(I) gauche conformation of $1,2$ -dibromoethane
(II) anti conformation of $1,2$ -dibromoethane
(III) trans- $1,4$ -dibromocyclohexane
(IV) cis- $1,4$ -dibromocyclohexane
(V) tetrabromomethane
(VI) $1,1$ -dibromocyclohexane

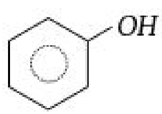
A) I and II
B) I and IV
C) II and V
D) I,IV and VI

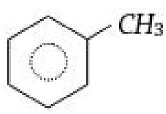
55. $\underset{(I)}{CH_2=CH-CH=CH-{}^{\oplus}NH_3;}$
$\underset{(II)}{{}^{\oplus}CH_2-CH=CH-{}^{\ominus}CH-{}^{\oplus}NH_3}$
$\underset{(III)}{{}^{\oplus}CH_2-CH=CH-CH=NH_3}$
Which of these structures is not a valid canonical structure ?
$\mathrm{CH_2=CH-CH=CH-{}^{\oplus}NH_3;{}^{\oplus}CH_2-CH=}$

A) I
B) II
C) III
D) none of these

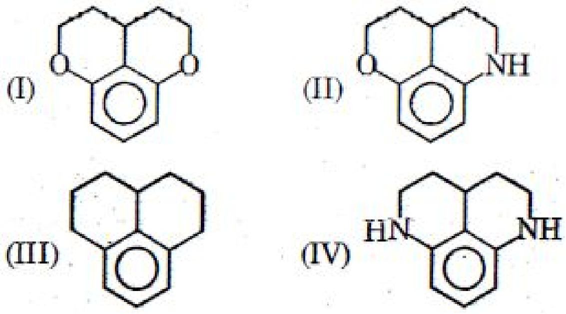
56. Which of the following compounds will be most easily attacked by an electrophile

A) 
B) 

C) 

D) 

57. In above compounds, decreasing order of rate of electrophilic aromatic substitution will be ?



A) $I > Il > III > IV$
B) $III > IV > II > I$
C) $IV > II > I > III$
D) $IV > I > II > III$

58. Match List - I with List - II

List - I (Pair of Compounds)	List - II (Isomerism)
An-propanol and Isopropanol	(I) Metamerism
BMethoxypropane and ethoxyethane	(II) Chain Isomerism
CPropanone and propanal	(III) Position Isomerism
DNeopentane and Isopentane	(IV) Functional Isomerism

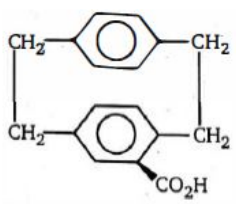
Choose the correct answer from the options given below:

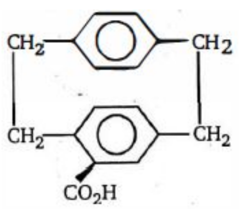
A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
B) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

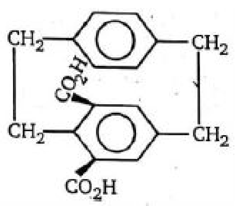
59. The number of possible structural alkynes with molecular formula, C_5H_8 is

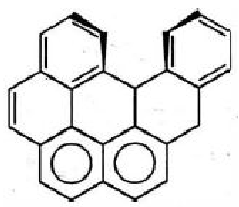
A) 2 **B)** 3 **C)** 4 **D)** 5

60. Which of the following compound is achiral ?

A) 

B) 

C) 

D) 

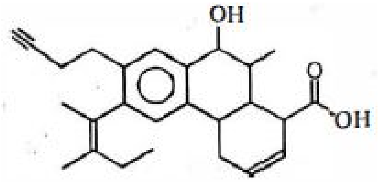
61. Predict stereochemistry of product when d and l -amine reacts with l-acid

A) Diastereomers **B)** Meso
C) Racemic **D)** Pure Enantiomer

62. $Ph - CH = NO_2H \xrightarrow[3days]{isomerises} \underset{(50\text{–}50\%)}{(x)}$, Isomer (x) is

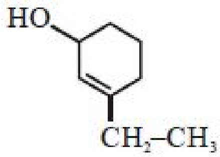
A) $Ph - NO - CH_2OH$
B) $Ph - CH_2 - NO_2$
C) $Ph - NH - CO_2H$
D) None

63. How many degrees of unsaturation are there the following compound ?



A) 6 **B)** 7 **C)** 10 **D)** 11

64. The $IUPAC$ name of following compound is

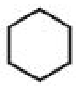
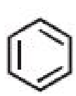


A) $1-$ Ethyl cyclohex $-1-$ en $-3-$ ol
B) $1-$ Ethyl $-3-$ hydroxy cyclohex $-1-$ ene
C) $3-$ Ethyl cyclohex $-2-$ en $-1-$ ol
D) $2-$ Ethyl $-6-$ hydroxy cyclohexene

65. The $IUPAC$ name of
$$\begin{array}{l} CH_2 - C = CH - CH_2OH \\ \;| \qquad\;\; | \\ Cl \qquad CH_3 \end{array}$$
is

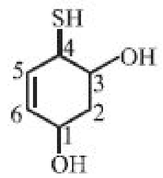
A) $4-$ Chloro $-3-$ methylbut $-2-$ en $-1-$ ol
B) $1-$ Chloro $-2-$ methylbut $-2-$ en $-4-$ ol
C) $4-$ Chloro $-1-$ hydroxy $-3-$ methylbut, $2-$ ene
D) $1-$ Chloro $-4-$ hydroxy $-2-$ methylbut $-2-$ ene

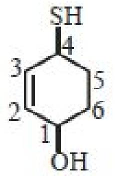
66. Identify the compound which is homocyclic, aromatic, and unsaturated ?

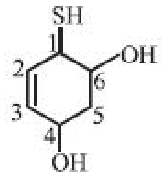
A)  **B)**  **C)** 

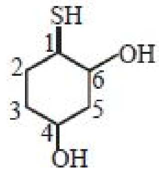
D) 

67. Select the structure with correct numbering for $IUPAC$ name of the compound

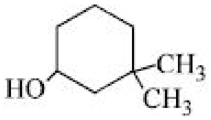
A) 

B) 

C) 

D) 

68. The $IUPAC$ name of the given compound is :



A) $1,1-$ dimethyl $-3-$ hydroxy cyclohexane
B) $3,3-$ dimethyl $-1-$ hydroxy cyclohexane
C) $3,3-$ dimethyl $-1-$ cyclohexanol
D) $1,1-$ dimethyl $-3-$ cyclohexanol

69. In the structure $$\begin{array}{c} CH_3 \\ | \\ {}^1H_3C{-}{}^2C{-}{}^3CH_2{-}{}^4CH_3 \\ | \\ CH_3 \end{array}$$ Which one is quarternary carbon atom.......

A) 1 B) 2 C) 3 D) 5

70. The compound which could not act both as oxidising as well as reducing agent is

A) SO_2
B) MnO_2
C) Al_2O_3
D) CrO

71. In the following reaction, $3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3$

A) Bromine is oxidised and carbonate is reduced
B) Bromine is reduced and water is oxidised
C) Bromine is neither reduced nor oxidised
D) Bromine is both reduced and oxidised

72. The number of moles of sodium sulphite (Na_2SO_3) needed to react with one mole of $KMnO_4$ in acidic solution

A) 0.8 B) 2.5 C) 1 D) 0.6

73. The correct set of oxidation number of N in NH_4NO_2 is

A) $-3,\ +5$
B) $+5,\ -3$
C) $-3,\ -3$
D) $-3,\ +3$

74. Assign A, B, C, D from given type of reaction. $I_2 + NaOH \longrightarrow NaI + NaOI$

A) for disproportionation reaction.
B) for comproportionation reaction.
C) for either intermolecular redox reaction or displacement reaction.
D) for either thermal combination redox reaction or thermal decomposition redox reaction.

75. In a chemical reaction, a reductant

A) Loses electron (s)
B) Gains electron (s)
C) loses and gains electrons both
D) No change in electrons

76. What is the molar solubility of :- $Ag_2CO_3(K_{sp} = 4 \times 10^{-13})$ in $0.1\,M\,Na_2CO_3$ solution?

A) 10^{-6}
B) 10^{-7}
C) 2×10^{-6}
D) 2×10^{-7}

77. Would gaseous HCl be considered as an Arrhenius acid

A) Yes
B) No
C) Not known
D) Gaseous HCl does not exist

78. An example for a strong electrolyte is

A) Urea
B) Ammonium hydroxide
C) Sugar
D) Sodium acetate

79. Which of the following substance is an electrolyte

A) Chloroform
B) Benzene
C) Toluene
D) Magnesium chloride

80. Which is generally true about ionic compounds

A) Have low boiling point
B) Have low melting point
C) Soluble in non polar solvents
D) Conduct electricity in the fused state

81. The addition of a polar solvent to a solid electrolyte results in

A) Polarization
B) Association
C) Ionization
D) Electron transfer

82. $\mathrm{PCl_5 \rightleftharpoons PCl_3 + Cl_3} \quad \mathrm{K_c} = 1.844$
3.0 $moles$ of $\mathrm{PCl_5}$ is introduced in a 1 L closed reaction vessel at 380 K. The number of moles of $\mathrm{PCl_5}$ at equilibrium is $.....\ \times 10^{-3}$. (Round off to the Nearest Integer)

A) 1500 B) 1292 C) 1400 D) 5123

83. The yield of product in the reaction, $A_2(g) + 2B(g) \rightleftharpoons C(g) + Q\,kJ$ would be higher at

A) low temperature and high pressure
B) high temperature and high pressure
C) low temperature and low pressure
D) high temperature and low pressure

84. For the reactions
$2NO + O_2 \rightleftharpoons 2NO_2\,;\,K_1$
$4NO + 2Cl_2 \rightleftharpoons 4NOCl\,;\,K_2$
$NO_2 + \frac{1}{2}Cl_2 \rightleftharpoons NOCl + \frac{1}{2}O_2\,;\,K_3$
where K_1, K_2, K_3 are equilibrium constants then K_3^2 equal to

A) $\sqrt{K_2/K_1}$ B) $\sqrt{K_1K_2}$
C) $\sqrt{K_2}/K_1$ D) $\frac{1}{K_1K_2}$

85. For the reversible reaction in equilibrium
$N_2(g) + O_2(g) \underset{k_2}{\overset{k_1}{\longleftrightarrow}} 2NO(g)$
$C_0 = Ce^{-2.1\times10^{-3}\,t}$ for the forward reaction and $C_0' = C'e^{-4.2\times10^{-4}\,t}$ for the backward reaction, hence K_c for the above equilibrium is

A) 5.0 B) 2.0 C) 0.5 D) 2

86. For the reversible reaction in equilibrium
$N_2(g) + O_2(g) \underset{k_2}{\overset{k_1}{\longleftrightarrow}} 2NO(g)$
$C_0 = Ce^{-2.1\times10^{-3}t}$ for the forward reaction and $C_0' = C'e^{-4.2\times10^{-4}t}$ for the backward reaction, hence K_c for the above equilibrium is :-

A) 5.0 B) 2.0 C) 0.5 D) 0.2

87. Total number of moles for the reaction $2HI \rightleftharpoons H_2 + I_2$. if α is degree of dissociation is

A) 1 B) $2-\alpha$
C) 2 D) $1-\alpha$

88. According to Le-chatelier's principle, an increase in the temperature of the following reaction will
$N_2 + O_2 \rightleftharpoons 2NO - 43,200\,kcal$

A) Increase the yield of NO
B) Decrease the yield of NO
C) Not effect the yield of NO
D) Not help the reaction to proceed in forward direction

89. The unit of equilibrium constant K for the reaction $A + B \rightleftharpoons C$ would be

A) $mol\ litre^{-1}$ B) $litre\ mol^{-1}$
C) $mol\ litre$ D) Dimensionless

90. Which of the following reactions is reversible

A) $H_2 + I_2 \rightarrow 2HI$
B) $H_2SO_4 + Ba(OH)_2 \rightarrow BaSO_4 + 2H_2O$
C) $NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl$
D) $Fe + S \rightarrow FeS$

Biology - (Zoology)

91. Match List I with List II

List I	List II
A. Pons	I. Provides additional space for Neurons, regulates posture and balance.
B. Hypothalamus	II. Controls respiration and gastric secretions.
C. Medulla	III. Connects different regions of the brain.
D. Cerebellum	IV. Neuro secretory cells

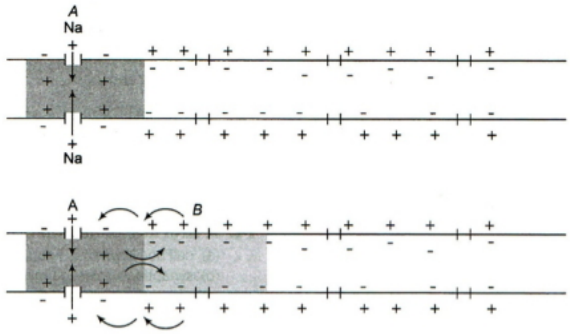
Choose the correct answer from the options given below :

A) $A-III, B-IV, C-II, D-I$
B) $A-I, B-III, C-II, D-IV$
C) $A-II, B-I, C-III, D-IV$
D) $A-II, B-III, C-I, D-IV$

92. Given is the diagrammatic representation of impulse conduction through an axon (at points A and B). View the diagram and arrange the steps of impulse conduction
I. The polarity of the membrane at site A is reversed and depolarized, i.e., the outer surface becomes negatively charged and the innerside becomes positively charged, generating nerve impulse
II. A stimulus causes disturbance to the membrane at site of A nerve fibre resulting in leakage of Na^{+ii} ions inside the nerve fibre

III. On the outer surface, current flows from site B to site A to complete the circuit of current flow. Hence, the polarity at the site is reversed, and an action potential is generated at site B. The impulse (action potential) generated at site A arrives at site B. The sequence is repeated along the length of the axon and consequently the impulse is conducted

IV. Immediately ahead, the axon ($e \cdot g.$, site B) membrane has a positive charge on the outer surface and a negative charge on its inner surface. As a result, a current flows on the inner surface from site A to site B



A) $I \rightarrow II \rightarrow IV \rightarrow III$
B) $II \rightarrow I \rightarrow III \rightarrow IV$
C) $II \rightarrow I \rightarrow IV \rightarrow III$
D) $I \rightarrow IV \rightarrow III \rightarrow II$

93. The system, responsible for providing an organized network of point to point connections for a quick coordination, is called

A) Endocrine system B) Circulatory system
C) Digestive system D) Neural system

94. In which of the following very simple neural organization is found ?

A) Insect B) Hydra C) Frog D) Human

95. Select the correct arrangement of neural organization, according to the increasing degree of complexity.

A) Lower invertebrates $\rightarrow$ Vertebrates $\rightarrow$ Insects
B) Lower invertebrates $\rightarrow$ Insects $\rightarrow$ Vertebrates
C) Vertebrates $\rightarrow$ Insects $\rightarrow$ Lower vertebrates
D) Vertebrates $\rightarrow$ Lower invertebrates $\rightarrow$ Insects

96. It is not found in or a part of myelinated nerve fibres

A) Schwann cell B) Node of ranvier
C) Nissl's granules D) Synaptic knob

97. Which of the following part contains centers, control respiration cardiovascular reflexes and gastric secretions

A) Cerebral aqueduct B) Pons
C) Cerebellum D) Medulla

98. The hormones that initiates ejection of milk, stimulates milk production and growth of ovarian follicles are respectively known as

A) $PRL, OT\ and\ LH$
B) $OT, PRL\ and\ FSH$
C) $LH, PRL\ and\ FSH$
D) $PRH, OT\ and\ LH$

99. Endocrine glands are

A) Ductless glands whose secretions pour directly into blood
B) Have ducts and pour their secretions into blood directly
C) Have ducts which straightway pour secretions into target organs
D) All of the above

100. Hypersecretion of growth hormone in adults does not cause further increase in height, because

A) epiphyseal plates close after adolescence
B) bones loose their sensitivity to growth hormone in adults
C) muscle fibres do not grow in size after birth
D) growth hormone becomes inactive in adults.

101. Endocrine glands produce or Action of endocrine glands is mediated through

A) Hormones B) Enzymes
C) Minerals D) Vitamins

102. Which of the following cell does not secrete hormone

A) Kupffer cell
B) Leydig cell
C) Lutein cell
D) Parafollicular cells of thyroid

103. Body co-ordination is exhibited by

A) Blood vascular system
B) Nervous system
C) Endocrine system
D) Nervous and endocrine system

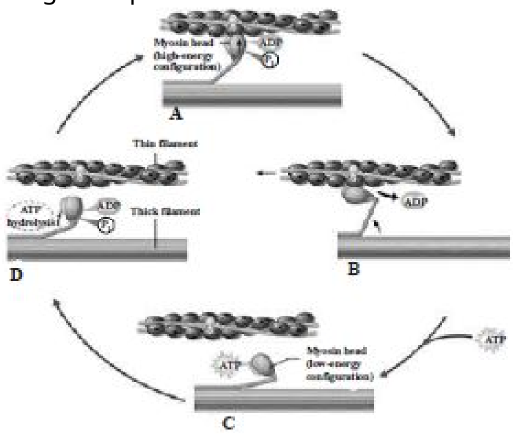
104. Endocrine glands

A) Do not possess ducts
B) Sometimes do not have ducts
C) Pour their secretion into blood through ducts
D) Always have ducts

105. Select the correct matching of the type of the joint with the example in human skeletal system:

A) Type of joint – Cartilaginous joint, Example – between frontal and parietal
B) Type of joint – Pivot joint, Example – between third and fourth cervical vertebrae
C) Type of joint – Hinge joint, Example – between humerus and pectoral girdle
D) Type of joint – Gliding joint, Example – between carpals

106. The given figure represents the cross bridge cycle in skeletal muscle. What does the step B in the figure represents ?



A) Attachment of myosin head to actin forming cross bridge.
B) Release of phosphate. Myosin changes shape to pull actin
C) Attachment of new ATP to myosin head. The cross bridge detaches
D) Splitting of ATP into ADP and Pi. Myosin cocks into its high energy conformation.

107. In the rib cage formation which is not used ?

A) Sternum
B) Ribs
C) Lungs
D) Thoracic vertebra

108. Muscles of the heart are

A) Voluntary striated
B) Voluntary smooth
C) Involuntary striated
D) Involuntary smooth

109. In human body which one of the following is anatomically correct?

A) Collar bones -3 pairs
B) Kidney -2 pairs
C) Cranial nerves -10 pairs
D) Floating ribs -2 pairs

110. The cup-shaped structure of pelvic girdle, the acetabulum in man is formed by

A) lium, ischium and pubis
B) Ilium, ischium and cotyloid
C) Ilium and ischium
D) Ilium and cotyloid

111. The pectoral and pelvic girdles and the bones of limb form

A) Axial skeleton
B) Appendicular skeleton
C) Visceral skeleton
D) Outer skeleton

112. Nitrogenous waste is excreted in the form of pellet or paste by :

A) Salamandra
B) Hippocampus
C) Pavo
D) Ornithorhynchus

113. Which of the following pairs is wrong?

A) Uricotelic -Birds
B) Ureotelic -Insects
C) Ammonotelic -Bony fishes
D) Ureotelic -Elephant

114. Human beings are

A) Uricotelic
B) Ureotelic
C) Ammonotelic
D) Both (b) and (c)

115. An advantage of excreting nitrogenous wastes in the form of uric acid is that

A) Uric acid can be excreted in almost solid form
B) The formation of uric acid requires a great deal of energy
C) Uric acid is the first metabolic breakdown products of acids
D) Uric acid may be excreted through the lungs

116. In the kidney, glucose is mainly absorbed in

A) Loop of Henle
B) Proximal convoluted tubules
C) Distal convoluted tubules
D) Bowman's capsule

117. Uricotelic mode of passing out nitrogenous wastes is found in

A) Reptiles and Birds
B) Birds and Marine fishes
C) Aquatic amphibians and Reptiles
D) Insects and amphibians

118. The most abundant, harmful and universal waste product of metabolism is

A) CO_2
B) Uric acid
C) H_2O
D) None of these

119. The opening of pulmonary vein is without valve because

A) It is a very small aperture
B) It has low blood pressure

C) Its opening is oblique
D) None of these

120. Sequence of electrical impulse in heart beat is

A) AVnode $\rightarrow$ pacemaker $\rightarrow$ auricles $\rightarrow$ ventricles
B) Ventricle $\rightarrow$ pacemaker $\rightarrow$ AV node $\rightarrow$ auricle
C) Pacemaker $\rightarrow$ atria $\rightarrow$ AV node $\rightarrow$ ventricle
D) Pacemaker $\rightarrow$ AV node $\rightarrow$ atria $\rightarrow$ ventricle

121. Grouping of ABO blood is based on the

A) Surface antigens present on $RBCs$
B) Surface lipids present on the cell membrane
C) Nature of all constituents
D) Nature of RBC and WBC

122. Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme........

A) Thrombin
B) Fibrinogen
C) Prothrombin
D) Thrombokinase

123. Match the following.

$Column\ -I$	$Column\ -II$
$(1)\ Neutrophil$	$(a)\ Inflammation$
$(2)\ Eosinophil$	$(b)\ 6-8\ \%$
$(3)\ Basophil$	$(c)\ Phagocytes$
$(4)\ Monocytes$	$(d)\ Resists\ infections$

A) $(1-d),(2-c),(3-a),(4-b)$
B) $(1-c),(2-d),(3-a),(4-b)$
C) $(1-a),(2-c),(3-b),(4-d)$
D) $(1-b),(2-a),(3-c),(4-d)$

124. Which of the following is a cell fragment?

A) Blood platelets
B) Bone cells
C) Lymphocytes
D) Leucocytes

125. Mark the odd one

A) Monocytes
B) Lymphocyte
C) Neutrophils
D) Erythrocytes

126. Match List I with List II :

List I	List II
$A.$ Expiratory capacity	$-$ Expiratory reserve volume $+$ Tidal volume $+$ Inspiratory reserve volume
$B.$ Functional residual capacity	$-$ Tidal volume $+$ Expiratory reserve volume
$C.$ Vital capacity	$-$ Tidal volume $+$ Inspiratory reserve volume
$D.$ Inspiratory capacity	$-$ Expiratory reserve volume $+$ Residual volume

Choose the correct answer from the options given below :

A) $A-III, B-II, C-IV, D-I$
B) $A-II, B-I, C-IV, D-III$
C) $A-I, B-III, C-II, D-IV$
D) $A-II, B-IV, C-I, D-III$

127. Process of exchange of O_2 from the atmosphere with ...A... produced by the cells is called ...B..., which is commonly known as ...C...
Choose the appropriate options for the blanks A, B and C to complete the given $NCERT$ statement

A) $A-$ H_2O, $B-$ breathing, $C-$ respiration
B) $A-$ O_2, $B-$ breathing, $C-$ respiration
C) $A-$ CO_2, $B-$ breathing, $C-$ respiration
D) $A-$ NO_2, $B-$ breathing, $C-$ respiration

128. What is correct at alveoli ?

A) High PO_2
B) High PCO_2
C) High H^+ concentration
D) Higher temperature

129. CO_2 is carried by haemoglobin as carbamino haemoglobin. This binding CO_2 with Hb is related to the partial pressure of ....

A) O_2, CO_2
B) Hb, RBC
C) O_2, Hb
D) CO_2, H_2O

130. Find out the false statement.

A) The overall exchange of gases between atmosphere, blood and cell is called respiration.
B) Excretory system and respiratory system involve in exchange of respiratory gases.
C) Cells need a continuous supply of oxygen to carry out activities for their existence.
D) Cellular respiration is the part of catabolism.

131. The normal shape of diaphragm is

A) Flat
B) Dome-like
C) Spherical
D) Cone-like

132. Oxy-haemoglobin dissociates into oxygen and deoxy-haemoglobin at

A) Low O_2 pressure in tissue
B) High O_2 pressure in tissue
C) Equal O_2 pressure inside and outside tissue
D) All times irrespective of O_2 pressure

133. In expiration condition, diaphragm becomes

A) Circular
B) Relaxed
C) Fully contracted
D) Expanded

134. Which of the following statement is correct

A) Breathing refers to chest movement and respiration to the another mechanism
B) Breathing means getting in of CO_2 in the lungs and O_2 out from the lungs
C) Respiration means the oxidation of food material
D) Breathing and respiration both are same

135. In which of the following animals, respiration occurs with out any respiratory organ

A) Fish B) Frog
C) Cockroach D) Earthworm

Biology - (Botany)

136. In coleoptile tissue, auxin is

A) Not transported because it is used where it is made
B) Transported by diffusion
C) Transported from the base to tip by osmosis
D) Produced by growing apices of stem, which migrate to the region of its action

137. Which of the following movements in plants is due to the increased concentration of auxin?

A) Movement of shoot towards the source of light
B) Nyctinasty
C) Movement of sunflower towards sun
D) All of the above

138. $I.$ Increased vacuolation
$II.$ Cell enlargement
$III.$ New cell wall deposition
Which of the above are the characteristics of phase of elongation?
Choose the correct option accordingly

A) I and II
B) II and III
C) I and III
D) I, II and III

139. Which hormone inhibit morphogenesis

A) ABA B) $2,4-D$
C) Jasmonic acid D) IBA

140. In order to increase the yield of sugarcane crop, which of the following plant growth regulators should be sprayed?

A) Ethylene B) Auxins
C) Gibberellins D) Cytokinins

141. Increase in the girth of plant is known as/done by

A) Primary growth
B) Apical meristem
C) Intercalary meristem
D) Secondary growth

142. Find out the correct statement(s).
(a) Growth in plants is internal/intrinsic and open ended
(b) Formation of cellular materials is called real or protoplasmic growth
(c) Plant growth is diffused only during the early embryonic stage

A) Only (a) & (b)
B) (b) only
C) Only (b) & (c)
D) $(a), (b)$ & (c)

143. $CH_2 = CH_2$ is mainly responsible for

A) Formation of internode
B) Formation of nodes
C) Ripening of fruits
D) Formation of internodes

144. If an etiolated stem could be first saturated with auxin by spraying and then exposed to a streak of light from one side, it will

A) Bend towards the light
B) Bend away from the light
C) Grow straight upwards
D) Be prevented from growing

145. Which of the following structures show unlimited growth in plants?

A) Leaves B) Flowers C) Fruits D) Roots

146. Cytokinins are formed in

A) Roots B) Leaves C) Fruits D) Stems

147. If a planter is interested in obtaining a good crop of tea leaves from a single plant, he should

A) Feed auxin to the plant through soil
B) Remove the apical bud of the main shoot and the branches
C) Cut off the tip of the plant and then apply auxins to the cut end
D) Supply auxin from the tip of the plant as well as through roots

148. Which two factors primarily affect the developmental phase of growth of plants

A) Light and temperature
B) Rainfall and temperature
C) Light and wind
D) Temperature and relative humidity

149. Where would you look for active cell division in plant

A) In the pith cells
B) In the cells of cortex
C) In the internodal region
D) At the tip of root and shoot

150. Plant growth in length is increased by

A) Apical meristem **B)** Lateral meristem
C) Dermatogen **D)** Periblem

151. In which part of mitochondria does ATP synthesis occur?

A) F_1
B) F_0
C) Cristae
D) Inner membrane of mitochondria

152. Choose the correct combination of A and B in accordance with the $NCERT$ text book.
The $NADH$ synthesised in ...A... is transferred into the mitochondria and undergoes oxidative ...B...

A) $A - EMP; B - carboxylation$
B) $A - ETS; B - phosphorylation$
C) $A - glycolysis; B - phosphorylation$
D) $A - TCAcycle; B - decarboxylation$

153. In alcohol fermentation

A) There is no electron donor
B) Oxygen is the electron acceptor
C) Triose phosphate is the electron donor, while acetaldehyde is the electron acceptor
D) Triose phosphate is the electron donor, while pyruvic acid is the electron acceptor

154. How many CO_2 are released by oxidation of 2 molecules of glucose?

A) 10 **B)** 8 **C)** 6 **D)** 12

155. RQ of anaerobic respiration is B

A) 1 **B)** 2
C) Less than 1 **D)** Infinite

156. The common immediate source of energy in cellular activity is

A) ADP **B)** ATP **C)** FAD **D)** NAD

157. How many ATP molecule are produced in Kreb's cycle

A) 36 **B)** 38 **C)** 40 **D)** 24

158. Which one is correct sequence in glycolysis

A) $G\,6 - P \rightarrow PEP \rightarrow 3 - PGAL \rightarrow 3 - PGA$
B) $G\,6 - P \rightarrow 3 - PGAL \rightarrow 3 - PGA \rightarrow 3 - PEP$
C) $G\,6 - P \rightarrow PEP \rightarrow 3 - PGA \rightarrow 3 - PGAL$
D) $G\,6 - P \rightarrow 3 - PGA \rightarrow 3 - PGAL \rightarrow PEP$

159. Kreb's cycle is also known as

A) Glyoxylate cycle **B)** EMP pathway
C) Citric acid cycle **D)** Glycolate cycle

160. Oxidative phosphorylation occurs during the process of

A) Protein synthesis
B) N_2 fixation
C) Respiration
D) Transpiration

161. The tissue of highest respiratory activity is

A) Meristems **B)** Ground tissue
C) Phloem **D)** Mechanical tissue

162. When ATP is converted into ADP, it releases

A) Electricity **B)** Energy
C) Enzymes **D)** Hormones

163. Carbon dioxide liberates during

A) Ascent of sap **B)** Transpiration
C) Photosynthesis **D)** Respiration

164. To a living organism which of the following has the greater amount of available energy per molecule

A) ATP **B)** ADP **C)** CO_2 **D)** H_2O

165. CO_2 is constantly released in respiration yet its level remains fairly uniform because

A) CO_2 is used as a buffer
B) CO_2 is utilized in photosynthesis
C) CO_2 is converted into carbonate rocks
D) CO_2 undergoes regular photolysis

166. The enzyme $RuBP$ carboxylase

A) Activity occurs in C_3 and C_4 plants
B) It present in inner thylakoid membrane
C) Is low temperature sensitive enzyme
D) Shows greater affinity for O_2 than for CO_2

167. Which of the following statements is not true regarding the C_4 plants?

A) They show kranz anatomy
B) Decarboxylation process occurs in bundle sheath cells
C) Granal chloroplast is present in bundle sheath cells.
D) PEPcase enzyme activity occurs in mesophyll cells

168. $NADPH$ is generated through

A) Anaerobic respiration
B) Cyclic photophosphorylation
C) Non-cyclic photophosphorylation
D) Glycolysis

169. In the chloroplast, the stroma lamellae lack the

A) $PS\ I$, $NADP$ reductase
B) $PS\ II$, $PS\ I$
C) $NADP$ reductase enzyme, $P700$
D) $NADP$ reductase, $PS\ II$

170. Which of the following is/are the raw material for photosynthesis?
$I.\ H_2O$ $II.\ CO_2$ $III.$ Light $IV.$ Chlorophyll
Choose the correct option

A) II, III and IV
B) I and IV
C) I, II and III
D) I, II, III and IV

171. Which is the evidence to show that O_2 is released in photosynthesis comes from water

A) Isotopic O_2 supplied as H_2O appears in the O_2 released in photosynthesis
B) Isolated chloroplast in water releases O_2 if supplied potassium ferrocyanide or some other reducing agent
C) Photosynthetic bacteria use H_2S and CO_2 to make carbohydrates
D) All the above

172. Aerobic atmosphere is maintained by

A) Prokaryotes B) Protists
C) Plants D) Fungi

173. What is the unique process which has supported life on this planet

A) N_2-fixation B) Photosynthesis
C) Protein synthesis D) Respiration

174. In C_3 plants, phosphoketopentose epimerage is required for converting

A) Ribose into ribulose
B) Xylulose to ribulose $1,5$ Di PO_4
C) Erythrose to xylulose
D) None of the above

175. No. of carboxylation in C_4 cycle is/are

A) 1 B) 2 C) 5 D) 3

176. Which of the following photosynthetic bacteria have both $PS-I$ and $PS-II$

A) Green sulphur bacteria
B) Purple sulphur bacteria
C) Cyanobacteria
D) Purple non-sulphur bacteria

177. The detectable end product of photosynthesis is

A) Glucose
B) Fructose $1,6$ diphosphate
C) Ribulose phosphate
D) Starch

178. Importance of photosynthesis is due to

A) Synthesis of food
B) Purification of atmosphere
C) Provided vast resources of energy
D) All the above

179. Photosynthesis is a

A) Exothermic process
B) Exergonic process
C) Anabolic process
D) Catabolic process

180. During formation of glucose which of the following is obtained from water

A) Oxygen
B) Hydrogen
C) Both (a) and (b)
D) None of the above

Mitoprep

Sapno Ki Udaan